

Homework 07: Chapter 10

EEC 3150

For each Python exercise, write a separate Python script. Submit all scripts on Blackboard under the appropriate assignment. Name your script files with the following format:

HW<assignment number>_Ex<exercise number>_<FirstInitial><LastName>.py

Example: HW01_Ex01_GWest.py

For short answer or math exercises, either do your work on a sheet of paper and upload a picture of it to Blackboard or submit some kind of text file or Word document to Blackboard. You can place all such exercises in a single document as long as you clearly label each exercise. These files should follow a similar naming convention as the Python scripts.

Exercise 1 (40 points, Python)

Create a class `Complex` which implements complex number arithmetic:

1. The class should have two attributes:
 - `r` (real part)
 - `i` (imaginary part)
2. The class should have these magic methods:
 - `__str__` should return a string in the following format `a+bi`
 - 1) make sure that if `b` is negative, it doesn't print `a+-bi`
 - 2) if either `a, b` are 0, don't print `0+bi` or `a+0i`, only print `0` if both `a, b` are 0 (you'll need some nested `if`'s in your method)
 - `__add__` should return a new `Complex` object that is the sum of the two operands
 - `__sub__` should return a new `Complex` object that is the difference of the two operands
 - `__mul__` should return a new `Complex` object that is the product of the two operands
 - `__truediv__` should return a new `Complex` object that is the quotient of the two operands
 - `__neg__` should return the negative of the complex number
 - `__eq__` should return whether two complex numbers are equal
 - `__ne__` should return whether two complex numbers are not equal
3. The class should have two additional methods:
 - `conj` (complex conjugate)
 - `mag` (the magnitude of the complex number via the Pythagorean Theorem)

For reference, here are some definitions for complex arithmetic. Let $x = a + bi$ and $y = c + di$ be two complex numbers.

The sum of x and y is:

$$\begin{aligned}x + y &= (a + bi) + (c + di) \\ &= (a + c) + (b + d)i\end{aligned}$$

The difference of x and y is:

$$\begin{aligned}x - y &= (a + bi) - (c + di) \\ &= (a - c) + (b - d)i\end{aligned}$$

The product of x and y is:

$$\begin{aligned}xy &= (a + bi)(c + di) \\ &= (ac - bd) + (ad + bc)i\end{aligned}$$

The quotient of x and y is:

$$\begin{aligned}\frac{x}{y} &= \frac{a + bi}{c + di} \\ &= \frac{(ac + bd)}{c^2 + d^2} + \frac{(bc - ad)}{c^2 + d^2}i\end{aligned}$$

Here's some examples of the class and methods:

```
63 for i in [-1,0,1]:
64     for j in [-1,0,1]:
65         print( Complex(i,j) )
66
67 print()
68
69 a = Complex(1,1)
70 b = Complex(0,1)
71 print(a)
72 print(b)
73 print(a+b)
74 print(a-b)
75 print(a*b)
76 print(a/b)
77 print(-a)
78 print(a==a)
79 print(a==b)
80 print(a!=a)
81 print(a!=b)
```

```
-1-1i
-1
-1+1i
-1i
0
1i
1-1i
1
1+1i
1+1i
1i
1+2i
1
-1+1i
1.0-1.0i
-1-1i
True
False
False
True
```